

Inter-City Freight Load Matching Marketplace

Requirements Table – Problem Statement #28

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A. Functional Requirements

ID	Requirement	Description	Priority	Acceptance Criteria	Rationale
FR-001	Post Freight Load	The system shall allow verified shippers to post freight shipments with cargo details, pickup and delivery locations, required vehicle type, and auction timeframe.	High	Pass: Shipment is created and visible to eligible carriers. Fail: Unverified shipper or incomplete details are rejected.	Helps shippers publish their loads.
FR-002	Submit Carrier Bid	The system shall allow verified carriers to submit competitive bids for available shipments	High	Pass: Bid is recorded and shipper is notified. Fail: Unverified carrier or	Allows carriers to bid and compete for loads.

		within the auction time.		expired auction bid is rejected.	
FR-003	Select Winning Bid	The system shall allow the shipper to review bids and select the best carrier based on price, rating, availability, and other factors.	High	Pass: Selected carrier gets confirmation and shipment becomes Assigned. Fail: Invalid or unavailable carrier cannot be selected.	Helps the shipper choose a suitable carrier.
FR-004	Track Shipment & POD	The system shall allow the shipper and carrier to track shipment milestones and allow the carrier to submit electronic proof of delivery (e-POD).	High	Pass: Milestones and valid e-POD are recorded and visible. Fail: Invalid or unsigned POD is rejected.	Provides shipment visibility and confirms delivery.
FR-005	Release Milestone Payment	The system shall release the milestone payment to the carrier automatically after successful verification of e-POD.	High	Pass: Payment is released and recorded. Fail: Payment remains held if verification fails.	Helps complete payment safely and on time.

B. Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	Milestone payments held in escrow shall be released automatically within 60 seconds of verified e-POD submission while maintaining secure processing.	High	Pass: At least 95% of valid releases finish within 60 seconds under peak-load testing and security tests pass. Fail: Target is missed.	Fast and secure payment builds trust.
NFR-002	Availability & Reliability	The marketplace shall have at least 99.5% monthly availability and shall not lose committed bids, shipment updates, or payment transactions during failures.	High	Pass: Monitoring shows at least 99.5% uptime and recovery tests show no loss of committed data. Fail: Target is missed or data is lost.	Freight operations need a reliable system.